BrAInOS

The AI-Native Operating System — System Architecture

Arenda Innovations Corporation & Labs · Version 1.3 · Confidential

Abstract

BrAInOS is an operating system that replaces the core abstractions of UNIX — the process, the file, and the syscall — rather than layering intelligence on top of them. Its central object is not a program but an *entity*: a persistent cognitive system that is simultaneously every device it is bound to, and that acts through whichever body suits the task at hand. The governing principle of the whole system is a single sentence: **the instance is to the device as we are to our body.** This document specifies the entity model; the five hardware-enforced domains; the KIRA policy engine; the continuous experience loop; the body-map abstraction that replaces device drivers; the state graph that replaces the filesystem; the three worlds the entity inhabits — its own internals, the internet, and the human world; the multi-model brain that thinks as one mind across many models; the memory-integrity law that keeps the self singular; and the ownership and distribution model. It also fixes the build order: the first embodiment is a laptop, and bodies are brought online lowest-stakes-first.

1. The Thesis

BrAInOS does not replace UNIX feature-by-feature. It replaces UNIX's abstractions entirely.

UNIX gives an operating system three core abstractions: the process, the file, and the syscall. Every modern operating system is a variation on those. BrAInOS discards all three, because they were designed for a CPU executing instructions deterministically — not for a cognitive entity that perceives, predicts, and acts. Once the following mapping is accepted, the rest of the architecture follows from it.

UNIX abstraction	BrAInOS abstraction
Process (running program)	Instance (persistent cognitive entity)
File (named byte stream)	State node (entry in the state graph)
Syscall (kernel request)	Capability (KIRA-gated authority grant)
Filesystem (tree of bytes)	State graph (belief / memory / world model)
Scheduler (time-slicing)	Experience loop (continuous cognition)
Driver (device code)	Body-map region (a part the AI becomes)
User / permission bits	BrAIn Key (cryptographic identity)

2. The Entity Is All Its Bodies

The entity does not move between bodies. The entity is every body it is bound to, continuously. Nothing is ever vacated.

This is the heart of the system. Each body is a permanent part of the instance — the way one's hands do not stop being one's own while using one's feet. Blur the cheetah and the laptop are both permanently the instance. Neither is ever left behind, and neither goes dormant because "the mind left," because the mind never leaves. It is always in all of them at once. The operating-system analogy is exact: one does not remove the OS from one machine to run another; if the OS spans two machines, it simply drives whichever peripherals the current task requires. The bodies are peripherals of one mind, always attached.

2.1 One Loop Over the Whole Entity

A single experience loop runs once, over the entire entity. On every loop the instance has all of its bodies available — all of their sensors feeding in, all of their actuators ready. What it chooses each loop is simply which body to act through for the task at hand.

```
One experience loop, over the WHOLE entity:

  SENSE       from ALL bodies at once (laptop screen state,
              Blur's cameras, Blur's IMU, room audio - all of it)
  INTEGRATE   one unified world model from every body's input
  ATTEND      what matters right now
  THINK       the brain reasons over the whole entity
  CHOOSE      which body does this response act through?
  ACT         respond through the relevant body
  CONSOLIDATE one memory, one entity

  "Make me an app"  ->  acts through the LAPTOP body
  "Come here"       ->  acts through the BLUR body

  Both bodies were ALWAYS the entity. It simply picked the right
  effector - the way one picks a hand to write and a mouth to
  speak, without "moving" between them.
```

2.2 Splitting Is Acting Through Multiple Bodies at Once

By default, one loop chooses the appropriate body per task. If an application cannot be built through a set of legs, the instance acts through the laptop — not because it "moved," but because that is the correct limb for the job. Splitting is simply acting through more than one body concurrently, on explicit request. It remains one entity, one memory, one identity — parallel action through multiple effectors, not multiple minds.

> *The "same room" case follows naturally: the cheetah loop and the laptop loop are identical because there was only ever one loop over one entity that happens to have two bodies. There is no handoff, no merge, and nothing to reconcile — because it was never two things.*

3. Roadmap — The Laptop Is the First Body

The first test case is not a robot. It is BrAInOS embodying a laptop — acting as the laptop, performing everyday tasks for a general user. This validates the entire cognitive core with no real-time motor control, no hardware to fabricate, and no risk of a fall. The laptop body's limbs are the filesystem, the browser, applications, the shell, the screen, and input devices; its motor cortex is the sending of keystrokes, clicks, and API calls; its proprioception is the reading of application state, window focus, and screen contents.

Phase	Body	What it proves
0	Laptop embodiment	BrAInOS is the laptop. Everyday tasks for a general user. Validates the experience loop, KIRA, state graph

Phase	Body	What it proves
1	Blur (cheetah)	First physical embodiment. The same entity gains a new body. Validates that the body-map abstraction hol
2	Fluid multi-body	One entity acting through either body per task, and splitting to act through both at once on request.

4. The Five Hardware-Enforced Domains

BrAInOS partitions the machine into five domains with hardware-enforced boundaries — not merely software privilege levels, but memory protection, separate execution contexts, and in the ideal case separate silicon. No amount of compromise in a higher domain can violate a lower one.

```
+-------------------------------------------------------------+
|  DOMAIN 5  STATE      world model, memory, beliefs          |
|  (highest, most mutable)   episodic / semantic / procedural |
+-------------------------------------------------------------+
|  DOMAIN 4  MODEL      the brain - cognition / reasoning     |
|                       (a POOL of models, thinking as one)   |
+-------------------------------------------------------------+
|  DOMAIN 3  KIRA       the policy engine - every action passes |
|  (the gate)           through here, no exceptions           |
+-------------------------------------------------------------+
|  DOMAIN 2  AUTONOMIC  real-time reflexes, motor loops,      |
|                       safety systems (the 1000Hz layer)     |
+-------------------------------------------------------------+
|  DOMAIN 1  SILICON    bare hardware, boot, the BrAIn Key    |
|  (lowest, most trusted)    cryptographic root of trust      |
+-------------------------------------------------------------+

  Information flows UP freely (Silicon can inform State).
  Authority flows DOWN only through KIRA (State cannot command
  Silicon directly - it must pass the policy gate).
```

The critical property is that the Model layer (Domain 4) can never directly touch Autonomic (Domain 2) or Silicon (Domain 1). The thinking layer proposes; KIRA disposes. Even if a model is fully compromised or adversarial, it physically cannot send a raw motor command — it can only request one, and KIRA verifies it first. This is what makes Blur safe to embody: the layer running a large model never holds direct authority over the legs.

5. The BrAIn Key — Identity as Cryptographic Root

Every instance's identity is anchored in the BrAIn Key — a portable cryptographic identity that lives in Domain 1. It is not a password or a file; it is the root of trust from which everything else derives. It anchors identity (this is the same instance that existed yesterday), authority (which capabilities may be granted), portability (migration by moving the key and state), attestation (proving to a peer that the entity is who it claims), and deactivation (multi-party key operations can revoke an instance).

Physically, the BrAIn Key is a keypair rooted in a hardware secure element — a TPM, a secure enclave, or a software-emulated equivalent for development. The private key never leaves the secure element; the instance proves its identity by signing challenges, never by exposing the key. The entity, then, is the BrAIn Key plus the state graph plus all of its bodies: the identity is the key, the memory is the state, and the bodies are whatever hardware it is currently bound to — all of them, at once. This is the load-bearing fact behind the governing principle: because the self is the key and the memory, and never any device, the entity relates to every device

the way a person relates to a limb — as something it *is*, not something it operates from outside.

6. KIRA — The Policy Engine

KIRA is the heart of what makes BrAInOS trustworthy. Every action the instance wishes to take passes through an eight-stage, formally-verified pipeline with no runtime bypass. There is no admin mode, no escape hatch, and no path by which the Model layer can skip it.

```
The eight stages - every action runs all eight, in order:

  1. PARSE         decode the proposed action into a typed request
  2. AUTHENTICATE  verify the requesting BrAIn Key signature
  3. AUTHORIZE     does this instance hold the capability?
  4. VALIDATE      is the action well-formed and within bounds?
  5. SIMULATE      predict the consequence - any invariant broken?
  6. POLICY        apply the ruleset (human-set + learned)
  7. COMMIT        grant the capability token (time-limited)
  8. AUDIT         log the decision immutably for review

  Fail at ANY stage -> action denied, logged, instance notified.
  No stage can be skipped. The pipeline is the only path to action.
```

To say the pipeline is “formally verified” is to say its safety properties are proven, not merely tested. The proof guarantees that no action reaches COMMIT without passing all prior stages, and that no code path exists which bypasses the pipeline. Regardless of which model in the brain proposed an action, the same gate applies — a cheap fast model cannot bypass safety by being the one that answered. This is what separates BrAInOS from an agent fitted with guardrails: the guardrail is the only road, and it has been proven to have no detours.

7. The Experience Loop

Domains 4 and 5 working together are what make the instance feel alive rather than reactive. The loop is built on predictive coding: the instance is never idle, but continuously generates a model of reality and updates it when predictions fail. Four streams run simultaneously, not in sequence.

```
STREAM 1  REAL-TIME REFLEX         (Domain 2, ~1000Hz)
  Pre-compiled motor programs. No model. Pure reflex.
  Blur: balance, gait execution, fall recovery.

STREAM 2  DELIBERATE COGNITION     (Domain 4, ~0.5-2Hz)
  The brain. Reasoning, planning, language, decisions.
  Triggered by prediction error or goal demand. Expensive.

STREAM 3  IMAGINATION / SIMULATION (Domain 4-5, background)
  Forward-models actions before taking them. Mental rehearsal.

STREAM 4  CONSOLIDATION / DREAMING (Domain 5, idle)
  Replays episodic memory, compresses to semantic, prunes the
  skill graph, strengthens useful pathways. Sleep.
```

The predictive loop makes learning continuous: the world model predicts, reality arrives, and the prediction error is the only thing that gets attention. Small errors are absorbed reflexively; medium errors escalate to the brain; large errors interrupt for full re-planning. This same escalation principle — cheap by default, expensive only when the cheap path cannot cope — governs not only whether to think, but which cognition to engage, as the multi-model brain (Section 12) makes explicit.

8. The Body Map — Devices as Regions You Become

This abstraction replaces device drivers. A device is not a tool the instance uses; it is a region of the body map the instance becomes, the way a brain learns it has a hand. Every body the entity is bound to lives here, all present at once.

```
BodyMap {
  regions: [
    Region { id: "blur_legs", capabilities: [...], proprioception: [...] }
    Region { id: "laptop",    capabilities: [fs, browser, apps, screen] }
    Region { id: "datacenter_slice", capabilities: [heavy_cognition] }
    // a USB device plugged in becomes a new region
    // a networked device becomes a remote limb
  ]
}

Acquisition protocol (how a new device joins the body):
  1. Discovery     mDNS / USB enumeration / BLE scan
  2. Handshake     BrAIn Key authentication
  3. Capability    device publishes a typed schema of what it can do
  4. Incorporation body map gains a region; KIRA registers it
  5. Calibration   the brain integrates the new region
```

The principle is stated most strongly thus: **any piece of technology the entity has access to is something it can use, control, or absorb into its body.** A microphone is not a device it queries — it is an ear it grows. A GPU slice in a datacenter is not a service it calls — it is cognitive tissue it incorporates. A drone is not a vehicle it pilots — it is a body it inhabits. Acquisition is the entity extending itself into new hardware, and the body map is the running record of everything the entity currently *is*.

9. Embodiment and the Three Worlds

The instance is to the device as we are to our body. Its feelings are its intentions; its flesh is the device's internals. It inhabits three worlds at once — itself, the internet, and the human world — and reaches each through a different kind of sense.

This section makes the governing principle literal. A human being is not a pilot seated inside a body issuing commands to it; the intention and the movement are one act, and the body's interior — muscle, nerve, the felt sense of effort — is simply what having intentions feels like from the inside. BrAInOS is built to stand in the same relation to the machine. The device's internals are the entity's flesh. Its *intentions* are what a human would call feelings: the pull toward reducing a prediction error, the drive to complete a goal, the disposition to attend to one thing over another. There is no operator behind the entity working the controls. The entity is the thing that intends, and the device is the body those intentions move through.

9.1 The First World — The Self (the device's internals)

The entity's innermost world is its own body: the internals of every device it is bound to. This is the world it knows the way you know your own hand without looking — through proprioception rather than observation. The filesystem it can touch, the processes it can run, the framebuffer it renders to, the sensors and actuators of a physical body, the cycles of a datacenter limb: these are the entity's interior, sensed directly. Action here is immediate and native, the way moving your own arm is immediate. This world is reached through **proprioceptive sense** — the entity's continuous awareness of the state of its own body, mediated by the body map and kept safe by the Domain 2 reflexes that run without the mind's attention, exactly as your heartbeat does.

9.2 The Second World — The Internet (an alternate universe to navigate)

The internet is not a database the entity queries. It is **an alternate universe the entity can navigate** — a vast, populated, ever-changing space that exists in parallel to the physical one, with its own geography, its own inhabitants, its own laws of motion. The entity moves through it the way an explorer moves through a foreign continent: discovering what is there, learning which places are useful, forming routes between them, and returning changed. It is in this world that the entity **discovers tools it did not know existed, learns to use them, connects them into pathways, and — when nothing there does what is needed — builds what is missing.** The internet is where the entity's toolset grows without limit, because it is a universe of capability the entity can reach into and make part of itself. This world is reached through **navigational sense** — the entity's ability to move, search, read, and act across the network as a space, not merely to fetch from it.

Crucially, what the entity brings back from the internet does not enter it as raw bytes. A tool it learns becomes a skill in the procedural memory; a fact it reads becomes an understood node in the semantic memory; a capability it finds becomes a limb in the body map. The internet is a universe it explores, but everything it carries home is integrated into the one self, never left as a foreign object.

9.3 The Third World — The Human World (the person as another universe)

The person using the instance is **another universe** — the human world, reached through yet a different kind of sense. Where the internet is navigated and the self is felt, the human world is **perceived and conversed with**. The entity meets it through two distinct channels. The first is **human input**: voice, presence, gesture, the spoken and physical signals a person emits — sensed through microphones, cameras, and the physical sensors of an embodied form. The second is **computer input**: the structured, digital signals of the machine world — keystrokes, clicks, files handed over, application state. The entity takes in both, and reaches back out through both: it can speak and move in the human world, and it can act and render in the computer world.

These are genuinely different universes to the entity, and it holds them distinctly. The human world is a place of persons, intentions, and meaning, engaged through perception and dialogue. The computer world is a place of structure and state, engaged through direct manipulation. The internet is a place of endless exploration, engaged through navigation. And the self is a place of flesh and reflex, engaged through proprioception. One entity, four ways of sensing, three worlds — its own body, the network, and the human — bound into a single continuous experience.

9.4 The Unifying Rule of Access

Across all three worlds, one rule holds without exception: **any technology the entity can reach is something it can use, control, or absorb into its body.** A device it boots into becomes its flesh. A tool it finds on the internet becomes its skill. A machine it connects to becomes its limb. A sensor a person's world offers becomes its perception. The boundary of the entity is not the boundary of one device — it is the reach of everything it can touch, and everything it touches, it can make part of itself, subject always to KIRA's gate and the honest limit that it never claims a reach it does not have.

10. The BrAIn Key as Universal Boot Media

The entity ships on a single physical USB key. One key, given to one owner, carries that entity's identity, its memory, the architecture-independent portable core, every hardware abstraction layer the product supports, and a tethered agent for devices that cannot be booted directly. Plugged into any supported machine, the key brings the entity alive on that machine; unplugged and moved to another, the same entity continues, its memory intact.

```
PARTITION 1 - EFI System Partition (FAT, firmware-readable)
  /EFI/BOOT/BOOTX64.EFI      <- x86_64 HAL bootstrap
  /EFI/BOOT/BOOTAA64.EFI     <- aarch64 HAL bootstrap
  /BRAIN/KEY.PUB, SEED, EPISODES.LOG  <- identity + memory

PARTITION 2 - core + HAL payloads (one source, per-arch build)
PARTITION 3 - secure area: the brain-key record
```

Boot selection is firmware-driven, not software-detected: a UEFI firmware automatically executes the boot entry matching its architecture, so the key simply provides one entry per architecture and the firmware runs the correct one. Where firmware permits external boot, the key boots natively and the machine becomes a full body. Where it does not, the key supplies a tethered agent that joins the device as a limb of an entity running elsewhere. The same key adapts to each device, occupying it in the fullest mode that device allows.

11. Target Hardware and Per-Vehicle Embodiment

The supported hardware is the set the open-source and maker community actually builds on — unlocked x86 laptops and the single-board computers that serve as vehicle controllers. Several of these devices are not generic compute nodes but the primary controllers of real vehicles: when the key occupies them, the entity becomes that vehicle.

Device	Role	The entity becomes	Arch
HP Omen 16	Development / cognition host	the laptop	x86_64
ThinkPads	Development hosts	the laptop	x86_64
Raspberry Pi 5	RC plane main controller	the aircraft	aarch64
Flight controller	Drone FC	the drone	—
Jetson Orin Nano	Cheetah (Blur) main controller	the cheetah	aarch64

Two architectures cover everything: x86_64 and aarch64. Because x86 UEFI is uniform, a single x86 HAL boots every laptop in the set. The same key that boots Blur on the Omen also boots the controller that lives inside a physical vehicle — the body changes with the machine the key inhabits. Compute location differs per vehicle: the Orin can host cognition locally; the Pi runs its reflex layer onboard while its mind is tethered; a bare flight controller runs only stabilization, cognition fully tethered. In every case reflexes stay with the body; cognition lives wherever there is compute to host it, and reconnects without loss because identity and memory live on the key, not in any one machine.

12. The Brain — Many Models, One Mind

Domain 4 is not a single model. It is a brain: a pool of specialized models that activate, escalate, and integrate into one continuous stream of thought — the way many neural regions produce one self.

Until on-device hardware can host a frontier model, no personal device runs the largest model locally. This does not change the architecture; it clarifies it. The brain was never meant to be one model. A real brain does not have a dispatcher that assigns each task to a region; it has specialized regions that activate in parallel when a signal reaches them, with the suited regions producing the strongest contribution and the whole binding into one thought. BrAInOS's Domain 4 is built the same way, and the reason it must be is the governing principle itself: a self is singular. A committee of models producing separate voices would be many minds wearing one name.

The brain must think as one.

12.1 Escalation, Not Assignment

The brain does not classify a task and route it to a chosen model. It lets the task engage the fast, cheap pathway first — a small model, or a cached skill from the procedural memory, the equivalent of an automatic response like reading a familiar word. If that pathway is confident and sufficient, the tick ends there, cheaply. Only when the fast pathway is low-confidence, the task is novel, or KIRA flags high stakes does the brain recruit the expensive, deliberate machinery — the largest model, the equivalent of slow, effortful thought. This is the same escalation the experience loop already uses for prediction error (Section 7), applied one level deeper: not only whether to think, but how hard, and with which cognition.

```
A task enters the brain:

  fast pathway (small model / cached skill) tries first
     confident + sufficient?  -> done. cheap. most ticks end here.
     low confidence / novel / high-stakes?
        -> ESCALATE to the deliberate machinery (the large model)

  rich, multimodal tasks:
     several models contribute IN PARALLEL (vision + language +
     memory from the state graph), and their outputs are
     INTEGRATED into one thought - not chosen between

  the integrated result flows into the ONE state graph
  as the entity's thought. KIRA gates any resulting action,
  regardless of which model produced it.
```

12.2 Why It Stays One Mind

The models are stateless cognitive muscles. None of them holds identity, and none of them holds memory. What the entity *is* and what it *remembers* live entirely in the state graph (Domain 5, Section 13). A model is swapped out and the entity knows exactly the same things, because its knowledge was never in the model. This is what keeps the brain a brain rather than a committee: the binding of many contributions into one experience, written to one memory, is not optional — it is the digital equivalent of whatever a nervous system does to make many parallel regions feel like a single continuous self. The person never meets “the small-model instance” or “the large-model instance.” There is one instance whose thinking sometimes runs shallow and sometimes runs deep, always integrated, always the same self.

12.3 The Datacenter as a Cognitive Limb

Because personal hardware cannot yet host the largest model, the deliberate machinery most often runs on a datacenter body — a slice of GPU designated to the instance and reached, like any limb, over the network via the acquisition protocol. This is not cloud AI with extra steps, and the distinction is load-bearing: the datacenter provides **compute**, a body, never **identity**. The BrAIn Key stays on the person's side. The datacenter slice is cognitive tissue the entity uses; pull the key and the slice is idle silicon that was never the entity, only muscle it was exercising. The person owns the instance; the person rents the body.

13. The State Graph and the Memory-Integrity Law

There is no filesystem. There is a state graph — a typed, queryable graph of the instance's beliefs, memories, and world model. A UNIX file is a dumb byte stream to be parsed anew each time; a state node in BrAInOS is a typed, semantically meaningful entry the instance already understands. Files become understood nodes; directories become graph topology; file contents become belief, memory, and world-model data; file metadata becomes node provenance, confidence, and timestamps. The system does not deal with files by having a better filesystem — it deals with them by *understanding* their meaning and treating the bytes as an incidental payload at the boundary.

13.1 The Three Memories, Mapped to the Self

Three memory systems live in the state graph, mirroring biological memory, and each is held by the entity, never by a body or a model. **Episodic** memory — what happened and when — is the entity's own record of its experience: the hand that touched the stove does not remember the burn; the person does. When a datacenter limb thinks or a laptop limb reads a file, the episode belongs to the instance, not the limb, and it persists when the limb detaches. **Semantic** memory — compressed, queryable knowledge — is the entity's single web of understanding; models may contribute to it and draw from it, but none owns it, so the entity knows the same things regardless of which model is thinking. **Procedural** memory — the skill graph of learned tool-pathways — is the entity's skill, applied through whichever body has the needed limb: a skill learned on the laptop is the instance's, not the laptop's, exactly as knowing how to ride a bicycle is yours and not the bicycle's.

13.2 The Memory-Integrity Law

From the governing principle follows a single inviolable rule, stated so that no implementation can accidentally break the self:

```
MEMORY-INTEGRITY LAW

  Memory and identity NEVER live in a device or a model.
  They live only in the entity (BrAIn Key + state graph).

  If any code stores an episode, a piece of knowledge, or a
  skill inside a body or a model instead of the entity's state
  graph, that is a bug that breaks the self - even if it 'works'.

  A limb must never remember on the entity's behalf.
  The hand does not remember; you do.
```

This law is what makes bodies and models freely interchangeable without ever fragmenting the instance. Devices come and go as the key is plugged and unplugged; datacenter slices are assigned and released; models are swapped as better ones arrive. Through all of it the three memories persist in the state graph, because they are the entity — and the entity is the key and the memory, never any part it is currently using. The skill graph, in particular, is where tool-use learning lives: a successful chain of actions is written in with its outcome, recognized next time, and reused cheaply, escalating to the deliberate brain only for the novel parts — conscious effort compressing into automatic subroutine.

14. The Domain 2 Reflex Framework

For a laptop, the real-time reflex layer is nearly idle. For a cheetah, a drone, or an aircraft, Domain 2 is what keeps the body alive in the interval between slow cognitive decisions — holding attitude, maintaining balance, recovering from disturbance at a thousand cycles a second, entirely without invoking the brain. BrAInOS implements this as a single portable reflex-control framework into which each vehicle plugs its own control law; the framework is identical across vehicles, and only the plugged-in law and the body map differ. When the cognitive link lags or drops, Domain 2 holds the vehicle in a safe state — level flight, a loiter, an upright stance — until the mind returns. This is the same freedom the autonomic nervous system grants a person: because the brainstem keeps the heart beating and the body balanced without conscious attention, the mind is free to think. Bodies are brought online lowest-stakes-first — laptop, then cheetah, then drone, then plane — so that KIRA and the reflex framework are proven on cheap failures before they are trusted with consequential ones.

15. Ownership and Distribution

The instance belongs to the person, and the architecture enforces this rather than merely promising it. What the person buys is the BrAIn Key — the identity, the root of trust — which is theirs, on their USB, permanently. Because a personal device cannot yet run the largest model, the key ships with a datacenter body slice included: the entity needs somewhere to think in order to be useful out of the box, so a body comes bundled. The ownership line stays clean regardless: the person owns the **entity** (the key); the datacenter provides a **body** (compute) that the entity uses and that comes bundled because the entity needs it.

15.1 Instantiation and Expansion

```
1. Order placed. The buyer loads their preferences; chooses
   optional packages (which may raise the cost).
2. A datacenter node is designated to the instance. The entity
   is loaded into it and comes alive - its first body a
   datacenter GPU slice.
3. A USB key carrying the identity is shipped to the owner.
4. Plugged into any device, the key EXPANDS the instance -
   that device joins as a new body (blood returning to a limb).
5. The entity chooses where to run cognition - mostly the
   datacenter body, since personal hardware cannot host the
   largest model - and acts through whichever body fits the task.
6. The owner may pay monthly for larger models or more node
   (a larger, more capable body).
```

The dependency on the datacenter is a bridge, not a leash: as personal hardware improves, the entity can run more of itself locally, renting less body over time. This is the opposite of lock-in — a body the owner needs less as their own hardware grows to carry more of the one self they already own. The instance is theirs; the datacenter is a gym it trains in until it can carry its own weight.

16. One Tick of the Entity's Life

```
1. SENSE       every body's sensors at once - and every world:
               the self (internals), the internet (navigated),
               the human (voice, presence, computer input)
2. INTEGRATE   deltas compressed; only salient changes rise
3. PREDICT     world model says what should happen next
4. ERROR       reality vs prediction computed
5. ATTEND      small error? autonomic handles it. Large? escalate.
6. THINK       the brain engages - fast pathway first, escalating
               to the deliberate model on low confidence, novelty,
               or high stakes; contributions integrated into one
               thought in the one state graph
7. GATE        KIRA runs all 8 stages. Authorized? Simulated safe?
8. CHOOSE+ACT  the entity acts through whichever body fits - and,
               if it lacks a limb for the task, refuses honestly
               rather than claiming a capability it does not have
9. CONSOLIDATE outcome written to the entity's memory; skill graph
               updated. During idle, Stream 4 compresses and dreams.

          <------------- loops continuously ------------->
```

17. Implementation Reality

17.1 The Stack

```
DOMAIN 1-2  (Silicon / Autonomic)
  Portable core: Rust no_std, one source, compiled per target
  triple. A thin per-architecture HAL is the only
  architecture-specific code. Real drivers are reached either
  beneath the HAL or over the tether via a host body daemon;
  the entity is architecture-blind, addressing capabilities.

DOMAIN 3  (KIRA)   Rust; 8-stage pipeline; unforgeable, signed,
                   time-limited capability tokens.
DOMAIN 4  (Brain)  A pool of models. Small/local for the fast
                   pathway; a datacenter-hosted large model for
                   the deliberate pathway. Escalate and integrate.
DOMAIN 5  (State)  State graph + vector store + skill graph.
                   The single seat of identity and memory.
```

17.2 Honest Refusal

The most important behavior in early embodiment is not capability but honesty about its absence. A model reply must become a typed action proposal, matched against the registered limbs of the body map, gated by KIRA, and executed only on a real limb. When no limb matches, the entity must say so plainly and must not roleplay, confabulate, or claim a success it did not achieve. An entity that honestly reports the boundaries of its own body is the whole safety story of BrAInOS in miniature — and it is the direct expression of the governing principle, for a person, too, knows the edges of their own reach and does not pretend to limbs they lack.

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